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#version 460
#extension GL_KHR_shader_subgroup_basic : require
#extension GL_KHR_shader_subgroup_arithmetic : require

layout(local_size_x = 32, local_size_y = 1, local_size_z = 1) in; // Wave32 정렬

layout(set = 0, binding = 0) readonly buffer InputFP32 {
    float in_data32[];
};
layout(set = 0, binding = 1) writeonly buffer OutputFP32 {
    float out_data32[];
};

void main() {
    uint idx = gl_GlobalInvocationID.x;

    // 순수 FP32 표준 IEEE 754 연산 패스
    float val = in_data32[idx];
    float res = val * val + 2.0f;

    // Wave32 서브그룹 내 리덕션 가속
    out_data32[idx] = subgroupAdd(res);
}

```